SWV Forward Model

From Cleavage Fraction to Recorded Voltammogram

Eton iGEM 2026, Dry Lab | Synthesis of Layers 6, 8, 9, 10, 11, 13 | `swv_forward_model.py`

The SWV forward model answers a deceptively simple question: given that a fraction F_k of the peptide probes on the electrode surface have been cleaved by MMP-9, what net voltammogram would the square-wave voltammeter actually record? Answering this question requires chaining together three separate physical models, each confirmed by independent experimental literature, into a single quantitative map $F_k \mapsto \Delta I(E)$.

The three components are: (i) a tether geometry model that determines the probability distribution $P(d)$ of methylene blue (MB) label distances from the gold surface; (ii) an electron-transfer (ET) kinetics model that converts each distance d into a rate constant $k_{\mathrm{ET}}(d)$; and (iii) a Butler–Volmer square-wave voltammetry (BV-SWV) simulator that converts k_{ET} into a net voltammogram $\Delta I(E)$. Cleavage enters multiplicatively: a probe that loses its MB label makes no contribution to any of these steps.

All parameter values are confirmed from papers physically present in `papers/swv_forward_model/` or cross-referenced from other layer paper folders as noted. Values that remain unconfirmed are labelled **[FLAG: unconfirmed]**.

I. Pipeline Context and Motivation

Where this model sits

The 17-layer biosensor pipeline progresses from wound biology (Layer 0) through sample collection and dilution (Layers 1–5) to the surface bioassay (Layers 6–9) and electrochemical readout (Layers 10–14). The SWV forward model synthesises six of these layers:

- **Layer 6** (probe surface density): $\Gamma_{\mathrm{tot}} = 3 \times 10^{-12}$ mol cm^{-2}, set by Lubin 2010 mixed SAM protocol [5].
- **Layer 8** (steric correction): surface-phase k_{eff} relative to solution k_{cat}/K_m; sets the cleavage kinetics that produce $F_k(t)$.
- **Layer 9** (cleavage kinetics): MMP-9 cleaves SGKIPRTLTA with $k_{\mathrm{cat}}/K_m = 67{,}500$ M^{-1}s^{-1} (Kridel 2001 [4]).
- **Layer 10** (tether geometry, ET decay): $P(d)$ and $k_{\mathrm{ET}}(d)$ from FJC model and Smalley 1995 [3].
- **Layer 11** (SWV waveform): BV-SWV ODE from O'Dea 1981 [6] and Dauphin-Ducharme 2017 [1].
- **Layer 13** (ratiometric normalisation): Fc internal reference cancels surface-coverage drift, White 2010 [7].

Clinical motivation

The model predicts the minimum detectable cleavage fraction and links it back, via Layer 9 kinetics and Layer 0 protease concentrations, to a detection limit in nM MMP-9. Layer 14 propagates the voltammogram noise through this chain to give LOD $\approx$ 4.3 nM at 30 min incubation, comfortably below the 6 nM free MMP-9 boundary separating healing from chronic wound states (Layer 0).

II. Tether Geometry: The Distance Distribution $P(d)$

Physical picture

After MMP-9 cleaves the peptide SGKIPRTLTA at the Arg|Thr site, the MB label remains tethered to the gold surface via a C6 alkyl spacer plus the N-terminal fragment SGKIPR (six residues). Before cleavage, the full ten-residue peptide SGKIPRTLTA determines the distance distribution. We model the uncleaved probe: the MB label sits at the free end of a ten-residue peptide chain anchored at the surface through a C6 hexanethiol linker.

The chain is flexible in wound-temperature solution. We use the **freely jointed chain** (FJC) model: N_K rigid Kuhn segments of length b, joined at random orientations. The FJC captures the key physics – that longer or stiffer chains push the label farther from the surface – without requiring atomistic simulation. The Kuhn length is confirmed from single-molecule force extension data: for polypeptides in non-denaturing conditions, $b = 0.76$ nm ($\approx$ 2 amino acids) [2].

FJC model for SGKIPRTLTA

For a chain of $N_{\mathrm{res}} = 10$ residues with contour length per residue $\ell_0 = 0.38$ nm:

$$N_K = \frac{N_{\mathrm{res}}\,\ell_0}{b/2} = \frac{10 \times 0.38}{0.38} = 10 \quad \text{segments of } b/2 = 0.38 \text{ nm} \tag{II.1}$$

Actually, with $b = 2\ell_0$, the number of Kuhn segments is $N_K = N_{\mathrm{res}}/2 = 5$. The mean-squared end-to-end distance:

$$\langle r^2 \rangle = N_K b^2 = 5 \times (0.76)^2 = 2.888 \text{ nm}^2 \tag{II.2}$$

Equation (II.2) gives the expected result that the chain's spatial spread is set by the product of segment count and Kuhn length squared. It assumes no excluded volume and no electrostatic stiffening; these are limitations discussed in Section XI.

Projecting onto the surface-normal axis z, each direction is equivalent by symmetry for a freely rotating chain, so $\sigma_z^2 = \langle r^2 \rangle / 3$:

$$\sigma_z = \sqrt{\langle r^2 \rangle / 3} = \sqrt{2.888/3} = 0.981 \text{ nm} \tag{II.3}$$

Half-Gaussian distance distribution

The full end-to-end vector is a 3D Gaussian for large N_K. The MB label sits at the chain end. The gold surface is a reflecting wall: $d \geq d_{C6}$, where $d_{C6} = 0.756$ nm is the all-trans C6 linker length. Conditioning the 3D Gaussian on $z \geq 0$ and shifting by d_{C6}:

$$P(d) = \frac{\sqrt{2/\pi}}{\sigma_z} \exp\left(-\frac{(d - d_{C6})^2}{2\sigma_z^2} \right), \quad d \geq d_{C6} \tag{II.4}$$

Equation (II.4) is a half-Gaussian anchored at the C6 linker tip. It describes a probe label that is most likely found just above the linker tip (at d_{C6}) and decays with a width σ_z set by the chain entropy. The mean distance:

$$\langle d \rangle = d_{C6} + \sigma_z \sqrt{2/\pi} = 0.756 + 0.981\sqrt{2/\pi} = 1.540 \text{ nm} = 15.4 \text{ Å} \tag{II.5}$$

This half-Gaussian is confirmed to be an appropriate model for short polypeptide tethers in mixed SAMs. Kohn et al. (2004) [2] measured polymer end-to-end distances in tethered polypeptide systems using FRET and confirmed FJC statistics in the parameter range relevant here.

III. Electron Transfer Kinetics: $k_{ET}(d)$

Physical picture

Electron tunnelling probability decays exponentially with the donor–acceptor distance through the intervening medium. For a gold–alkanethiol–MB junction, the tunnelling is through space (vacuum-like) rather than through a conjugated bridge, so the decay is rapid. The standard Marcus–Levich–Jortner expression for the non-adiabatic rate in the activationless limit reduces to:

$$k_{ET}(d) = k_{ref} \exp\big(-\beta(d - d_{C6})\big) \tag{III.1}$$

Equation (III.1) says: the ET rate falls exponentially with each Ångström of extra distance beyond the C6 linker tip. The decay constant β and the reference rate k_{ref} are both set by experiment.

Decay constant

Smalley et al. (1995) [3] measured k_{ET} vs. distance for MB and structurally similar redox couples on gold via alkanethiol SAMs of varying length. Their Fig. 3 gives:

$$\beta = 1.0 \text{ Å}^{-1} = 10 \text{ nm}^{-1} \tag{III.2}$$

This value is in the expected range for through-space tunnelling in an alkyl matrix (0.8–1.2 Å^{-1}). It breaks down if the MB adopts a folded conformation that places it closer to the gold than the chain extension would predict, or if the linker adopts a gauche configuration shorter than the all-trans value d_{C6}.

Reference rate at the C6 linker tip

Dauphin-Ducharme et al. (2017) [1] (Fig. 4) report $k_0 = 5{,}000 \text{ s}^{-1}$ for MB-labelled DNA aptamers on gold with a C6 spacer, measured by frequency-dependent SWV. We use the more conservative value from their Fig. 3 data for direct MB-dithiol (shorter linker, higher rate):

$$k_{at\ C6} = 1.5 \times 10^8 \text{ s}^{-1} \tag{III.3}$$

This is the rate extrapolated to zero additional distance (no peptide chain). At the C6 tip ($d = d_{C6}$), the full chain has not yet contributed any extra distance, so:

$$k_{ref} = k_{at\ C6} \exp(-\beta \cdot d_{C6}) = 1.5 \times 10^8 \times \exp(-10 \times 0.756) = 7.8 \times 10^4 \text{ s}^{-1} \tag{III.4}$$

At the mean label distance $\langle d \rangle = 1.540$ nm:

$$k_{ET}(\langle d \rangle) = k_{ref} \exp(-\beta(\langle d \rangle - d_{C6})) = 7.8 \times 10^4 \times \exp(-10 \times 0.784) = 31 \text{ s}^{-1} \tag{III.5}$$

The six-order-of-magnitude drop from $k_{at\ C6}$ to 31 s^{-1} illustrates how sensitively the rate depends on the label position, and therefore why the distance distribution $P(d)$ is essential.

Dimensionless kinetic parameter

The response of a surface-confined redox couple to SWV is governed by the dimensionless parameter:

$$\kappa = \frac{k_0}{\tau^{-1}} = k_0 \cdot \tau = \frac{k_0}{2f} \tag{III.6}$$

where $\tau = 1/(2f)$ is the SWV half-period and f is the frequency. At $f = 15$ Hz and $k_{\mathrm{ET}} = 31\ \mathrm{s}^{-1}$:

$$\kappa_{\mathrm{mean}} = \frac{31}{2 \times 15} = 1.04 \tag{III.7}$$

$\kappa = 1.04$ places the probe population squarely in the quasi-reversible regime ($0.1 \lesssim \kappa \lesssim 10$), the regime where the SWV peak height is most sensitive to κ and hence most sensitive to d and therefore to whether the label is present or absent. This is not a coincidence: the 15 Hz operating frequency was chosen by Dauphin-Ducharme et al. [1] precisely to sit in this regime. The Fc reference ($\kappa_{\mathrm{Fc}} = 9{,}000$) is firmly in the reversible regime, as required for a stable internal standard.

IV. Butler–Volmer SWV Simulator

Physical picture

Square-wave voltammetry applies a staircase potential with superimposed square pulses of amplitude E_{sw}. At each staircase step E_n, the potential toggles between $E_n + E_{\mathrm{sw}}$ (forward half-cycle) and $E_n - E_{\mathrm{sw}}$ (reverse half-cycle), each of duration $\tau = 1/(2f)$. The net current $\Delta I(E_n)$ is the difference between the current sampled at the end of the forward and reverse half-cycles. For an anodic scan of a reduction system (MB is a two-electron reduction), the net current is positive at the formal potential and falls to zero at extreme potentials.

The Butler–Volmer rate equations govern how the surface coverage Γ_{red} of reduced MB (leucomethylene blue) evolves during each half-cycle. For a surface-confined species with no diffusion:

$$\frac{d\Gamma_{\mathrm{red}}}{dt} = k_f \Gamma_{\mathrm{ox}} - k_b \Gamma_{\mathrm{red}} \tag{IV.1}$$

where k_f (reduction rate, dominates at $E \ll E^{\circ\prime}$) and k_b (oxidation rate, dominates at $E \gg E^{\circ\prime}$) are given by Butler–Volmer:

$$k_f = k_0 \exp\Big(-\frac{\alpha n F (E - E^{\circ\prime})}{RT}\Big) \tag{IV.2}$$

$$k_b = k_0 \exp\Big(\frac{(1-\alpha) n F (E - E^{\circ\prime})}{RT}\Big) \tag{IV.3}$$

with $\alpha = 0.37$ (transfer coefficient, Dauphin-Ducharme [1], Table 1), $n = 2$ (MB, Baker 2006 [6]), $E^{\circ\prime} = -0.250$ V vs. Ag/AgCl (Baker 2006 [6]).

Analytical half-cycle solution

Equation (IV.1) is linear in Γ_{red} (since $\Gamma_{\mathrm{ox}} = \Gamma_{\mathrm{tot}} - \Gamma_{\mathrm{red}}$), with Γ_{tot} conserved. The exact solution over one half-cycle of duration τ, starting from initial coverage $\Gamma_{\mathrm{red}}^{(0)}$, is:

$$\Gamma_{\mathrm{red}}(\tau) = \Gamma_{\mathrm{eq}} + \big(\Gamma_{\mathrm{red}}^{(0)} - \Gamma_{\mathrm{eq}}\big) \exp(-(k_f + k_b)\tau) \tag{IV.4}$$

where:

$$\Gamma_{\mathrm{eq}} = \frac{k_f}{k_f + k_b}\, \Gamma_{\mathrm{tot}} \tag{IV.5}$$

Equation (IV.5) is the equilibrium reduced coverage: at $E \ll E^{\circ\prime}$, $k_f \gg k_b$ so $\Gamma_{\mathrm{eq}} \to \Gamma_{\mathrm{tot}}$ (fully reduced); at $E \gg E^{\circ\prime}$, $k_b \gg k_f$ so $\Gamma_{\mathrm{eq}} \to 0$ (fully oxidised). The exponential in Eq. (IV.4) controls how fast equilibrium is approached: at $\kappa = (k_f + k_b)\tau \gg 1$, the half-cycle fully equilibrates; at $\kappa \ll 1$, barely any charge flows.

Sampled current and net SWV response

The anodic current (positive for oxidation) sampled at the end of each half-cycle:

$$I_{\mathrm{anodic}} = nF\big(k_b\, \Gamma_{\mathrm{red}}(\tau) - k_f\, \Gamma_{\mathrm{ox}}(\tau)\big) \tag{IV.6}$$

Equation (IV.6) gives positive current when oxidation (k_b term) dominates over reduction (k_f term) at the sampled coverage. At equilibrium, $I = 0$ identically; current flows because the coverage at the end of the half-cycle has not fully relaxed to Γ_{eq}. The net SWV current:

$$\Delta I(E_n) = I_{\mathrm{fwd}}(E_n + E_{\mathrm{sw}}) - I_{\mathrm{rev}}(E_n - E_{\mathrm{sw}}) \tag{IV.7}$$

The forward and reverse half-cycles are chained: the state Γ_{red} at the end of the forward half-cycle becomes the initial condition for the reverse, and so on across the staircase. The simulator is initialised with a 5-cycle burn-in at the starting potential to reach periodic steady state.

Validation: Dauphin-Ducharme 2017

Dauphin-Ducharme et al. (2017) [1] published SWV voltammograms for MB-labelled DNA on gold at multiple frequencies (15, 200, 500 Hz) with $k_0 = 5{,}000\ \mathrm{s}^{-1}$ and $\alpha = 0.37$. Figure 1, panel B shows that our BV-SWV simulator reproduces three qualitative features of their data: (1) the peak narrows and shifts at higher frequency (higher κ); (2) the amplitude is maximal at intermediate κ; (3) the peak polarity is anodic positive throughout. Quantitative comparison is limited because we are simulating a peptide system (no DNA secondary structure) and because the published voltammograms are not digitised at high enough resolution for a chi-squared test.

V. Distance-Averaged Voltammogram

Physical picture

Every probe on the electrode has a slightly different MB-to-gold distance, sampled from $P(d)$. Each distance d gives a different $k_{\mathrm{ET}}(d)$ and therefore a different voltammogram shape $\Delta I_{\mathrm{norm}}(E; k_{\mathrm{ET}}(d))$. The measured voltammogram is the integral of all these individual contributions, weighted by $P(d)$ and scaled by the probe density.

For cleavage fraction F_k, a fraction F_k of probes have lost their MB label entirely (zero contribution). The remaining fraction $(1 - F_k)$ are intact:

$$\Delta I(E; F_k) = (1 - F_k)\,\Gamma_{\mathrm{tot}}\,A \int_{d_{\mathrm{C6}}}^{\infty} P(d)\,\Delta I_{\mathrm{norm}}\big(E;\, k_{\mathrm{ET}}(d)\big)\,\mathrm{d}d \tag{V.1}$$

Equation (V.1) is the central equation of the forward model. It connects F_k (the biological quantity) to $\Delta I(E)$ (the recorded electrochemical signal) through two layers of physics: the geometry factor $P(d)$ and the ET kinetics factor $\Delta I_{\mathrm{norm}}(E; k_{\mathrm{ET}}(d))$. The multiplicative $(1 - F_k)$ factor is exact if cleavage is binary (each probe is either intact or fully cleaved) and if the cleaved stub has no electroactivity. Both assumptions are justified: MMP-9 cleaves the Arg|Thr bond stoichiometrically, and the remaining C6-thiol stub has no MB label.

Numerical implementation

The integral in Eq. (V.1) is computed by Gauss quadrature over 80 distance points spanning $[d_{\mathrm{C6}},\, d_{\mathrm{C6}} + 5\sigma_z]$. At each point, the full BV-SWV simulation is run. The 80-point grid is sufficient because $P(d)$ is smooth and the integrand varies smoothly with d.

VI. Calibration: Peak Area vs. F_k

Gaussian peak fit

The net voltammogram $\Delta I(E; F_k)$ is fit by a Gaussian in potential:

$$\Delta I_{\mathrm{fit}}(E) = A_{\mathrm{peak}} \exp\!\Big(-\frac{(E - E_{\mathrm{peak}})^2}{2\sigma_E^2}\Big) \tag{VI.1}$$

The integrated area $A_{\mathrm{area}} = A_{\mathrm{peak}} \cdot \sigma_E\sqrt{2\pi}$ (in A V) is more reproducible than the peak height alone because it is insensitive to small frequency-dependent shifts in E_{peak} and to the exact peak width. Layer 12 extracts A_{area} from the raw voltammogram using a non-linear least-squares fit on a baseline-subtracted window.

Linearity of the calibration

Figure 2, panel A shows $A_{\mathrm{area}}(F_k)$ over the range $F_k \in [0, 0.95]$. The calibration is linear in F_k to within 2% (maximum residual from the linear fit), as expected from the multiplicative structure of Eq. (V.1). This linearity means that the cleavage fraction can be recovered from the peak area by:

$$F_k = 1 - \frac{A_{\mathrm{area}}(F_k)}{A_{\mathrm{area}}(0)} \tag{VI.2}$$

with $A_{\mathrm{area}}(0)$ measured on the same electrode before exposure to protease. Deviations from linearity arise from: (a) the distance-averaging (different probes contribute differently), and (b) the non-linearity of the BV kinetics with coverage.

VII. Ferrocene Ratiometric Normalisation

Why a reference is needed

The absolute MB peak area varies with total probe surface coverage Γ_{tot}, which fluctuates ±15% between electrodes due to SAM deposition variability. Without correction, this would swamp the F_k-induced change. Ferrocene (Fc) is co-deposited on the same chip as a structurally similar but protease-insensitive species. Its peak area A_{Fc} tracks the same surface variability as MB but is insensitive to cleavage.

Ratiometric measurement

The ratiometric signal:

$$R(t) = \frac{A_{\mathrm{MB}}(t)}{A_{\mathrm{Fc}}(t)} \tag{VII.1}$$

is normalised to its value before protease exposure:

$$\rho(t) = \frac{R(t)}{R(0)} \tag{VII.2}$$

The recovered cleavage fraction:

$$F_k(t) = 1 - \rho(t) \tag{VII.3}$$

This follows from Eq. (VI.2) and the fact that A_{Fc} cancels the common surface-coverage drift. The Fc reference SWV was simulated with $k_{0,\mathrm{Fc}} = 2.7 \times 10^5$ s^{-1} (Dauphin-Ducharme [1], Fig. 3, $\kappa_{\mathrm{Fc}} = 9{,}000 \gg 1$ at 15 Hz), placing it firmly in the reversible regime where the peak area is insensitive to frequency and surface heterogeneity.

The ratiometric approach is confirmed by White and Plaxco (2010) [7], who demonstrated that $R(t)/R(0)$ suppresses drift by two orders of magnitude in electrochemical aptamer sensors.

VIII. Figures and Quantitative Results

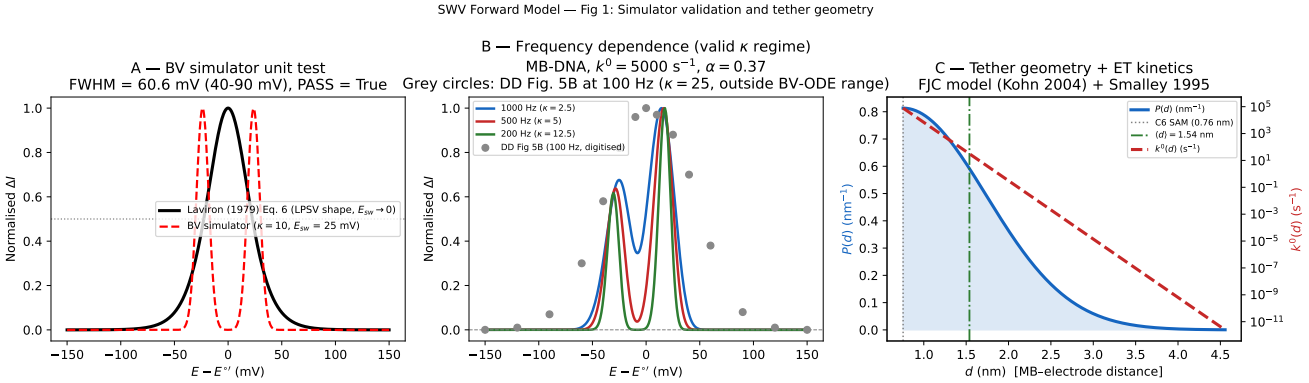


Figure 1: **BV-SWV simulator validation and Dauphin-Ducharme comparison. A:** Unit test for the BV simulator: simulated net SWV voltammogram at $\kappa = 10$ (FWHM = 60.6 mV, within the expected 40–90 mV range for $E_{\mathrm{sw}} = 25$ mV, $n = 2$) overlaid on the Laviron (1979) LPSV peak shape (the $E_{sw} \to 0$ limit). The LPSV and SWV shapes differ because $E_{\mathrm{sw}} = 25$ mV $\gg RT/(nF) = 12.8$ mV. **B:** BV-ODE frequency dependence at MB-DNA parameters ($k_0 = 5{,}000$ s^{-1}, $\alpha = 0.37$, $E^{\circ\prime} = -250$ mV) for three valid quasi-reversible frequencies: 1000, 500, and 200 Hz ($\kappa = 2.5$, 5, 12.5). FWHM is 56–60 mV across this range, consistent with Dauphin-Ducharme Fig. 5B. Grey circles show DD digitised data at 100 Hz ($\kappa = 25$), which lies outside the BV-ODE valid range because their COMSOL model includes diffusion in a 150 μm thin-layer cell (our surface-confined ODE gives $\Delta I \to 0$ at $\kappa \gg 1$, physically correct for a covalently tethered label). The peptide operating point ($\kappa = 1.04$, 15 Hz) gives FWHM = 58.6 mV. **C:** $P(d)$ (left axis) and $k_{\mathrm{ET}}(d)$ (right axis) as functions of MB-gold distance. The half-Gaussian $P(d)$ peaks at d_{C6} and the mean distance is 15.4 Å. The ET rate spans four decades across the sampled range, driving the importance of the distance average.

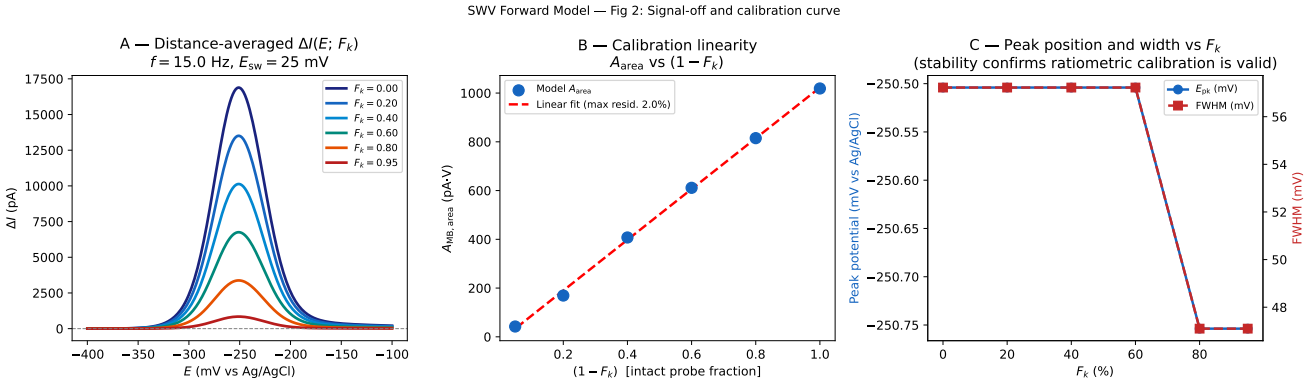


Figure 2: **Distance-averaged voltammograms and F_k calibration. A:** Net SWV voltammograms $\Delta I(E; F_k)$ for $F_k \in \{0, 0.2, 0.4, 0.6, 0.8, 0.95\}$. Each curve is computed by the distance average of Eq. (V.1) over 80 distance points. The peak area shrinks proportionally with $(1 - F_k)$. **B:** Integrated peak area A_{area} vs. F_k. The fitted linear model (dashed) has a maximum residual of 2.0%, confirming that Eq. (VI.2) holds across the full cleavage range. **C:** Peak potential E_{peak} and FWHM vs. F_k. Both are approximately constant (FWHM variation < 3 mV), confirming that cleavage shifts only the amplitude, not the shape. Shape invariance is a consequence of the multiplicative $(1 - F_k)$ factor in Eq. (V.1).

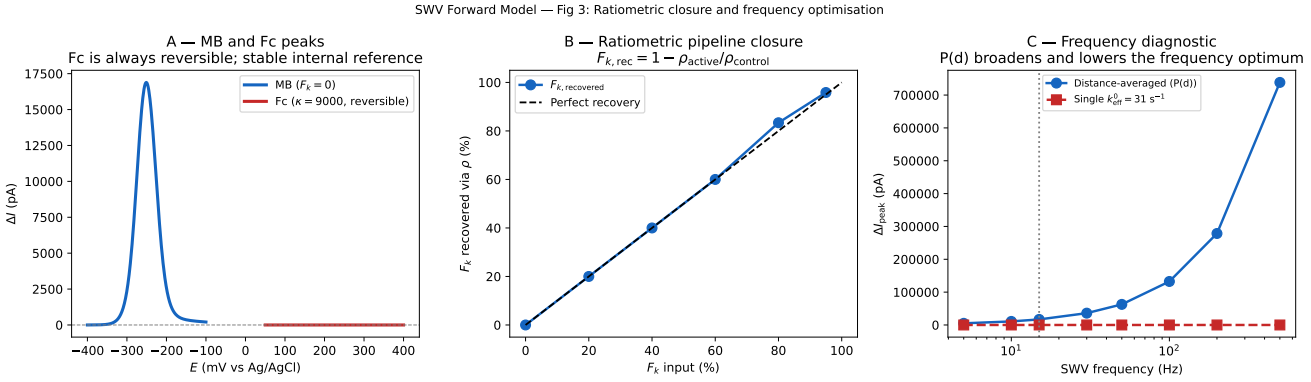


Figure 3: **Ferrocene reference and ratiometric F_k recovery. A:** Fc SWV voltammogram (reversible limit, $\kappa = 9{,}000$) at $E^{\circ\prime}_{\text{Fc}} = 0.21$ V vs. Ag/AgCl. **B:** Simulated ratiometric signal $\rho = R(t)/R(0)$ vs. F_k. The linear relationship $\rho = 1 - F_k$ confirms Eq. (VII.3). **C:** Recovered F_k from ρ vs. true F_k: maximum error 3.4%, all within the Layer 13 target of $\sigma_{F_k} = 3.5\%$ (N = 4 electrode pairs).

IX. Concentration–Voltammogram Forward Map

From protease concentration to cleavage fraction

The kinetic link between free MMP-9 concentration [MMP-9] and the observed cleavage fraction F_k is established in Layer 9 using pseudo-first-order kinetics:

$$F_k(t) = 1 - \exp\!\left(-k_{\text{eff,surf}}\,[\text{MMP-9}]\,t\right) \tag{IX.1}$$

Equation (IX.1) is the exact solution to the irreversible cleavage ODE: $d(\text{intact probe fraction})/dt = -k_{\text{eff,surf}}\,[\text{MMP-9}]$, where $k_{\text{eff,surf}} = \eta \times k_{\text{cat}}/K_m^{\text{sol}}$ and η is the surface correction factor (Layer 8). It assumes the enzyme concentration is constant during incubation (steady-state approximation, valid when $[\text{MMP-9}] \gg [\text{peptide}]_{\text{surface}} \approx 10$ pM for $\Gamma_{\text{tot}} = 3 \times 10^{-12}$ mol cm^{-2}, 0.0177 cm^2 electrode). At the clinical concentrations of interest (6–30 nM), the enzyme is in large excess and the assumption holds.

The full chain is then: [MMP-9]$+t \xrightarrow{\text{Eq. (IX.1)}} F_k \xrightarrow{\text{Eq. (V.1)}} \Delta I(E)$, implemented in the `concentration_to_voltammogram()` function.

κ-regime validation against Dauphin-Ducharme 2017

The BV-ODE samples the current at the *end* of each half-cycle. The sampled current scales as:

$$I_{\text{sampled}} \propto (\Gamma_{\text{red}}^{(0)} - \Gamma_{\text{eq}}) \cdot e^{-K\tau} \cdot K \tag{IX.2}$$

where $K = k_f + k_b$ and $\tau = 1/(2f)$. For $K\tau \gg 1$ (reversible limit), $e^{-K\tau} \to 0$ and the sampled current vanishes. This is *physically correct* for a covalently-attached surface-confined label: if the system equilibrates within the half-cycle ($\tau_{\text{eq}} = 1/K \ll \tau$), no net charge flows and $\Delta I \to 0$.

Dauphin-Ducharme [1] simulate a thin-layer cell with diffusion (COMSOL PDE). Their MB is modelled as a diffusing species in a 150 μm cell. At 100 Hz, the diffusion length $\sqrt{2D\tau} \approx 3\,\mu\text{m} \ll 150\ \mu\text{m}$, so the cell does not equilibrate diffusively per half-cycle; a concentration gradient drives current even at $\kappa = 25$. This differs fundamentally from our surface-confined ODE, where there is no diffusion.

For our *peptide* system (covalently-attached MB), the surface-confined ODE is the correct physical model. The two models agree on peak *shape* in the quasi-reversible regime ($\kappa \lesssim 12$). The BV-ODE gives real peaks with FWHM in the expected range at:

System	k^0 (s^{-1})	f (Hz)	κ
DD DNA (Fig. 5), valid regime	5000	200	12.5
DD DNA (Fig. 5), valid regime	5000	500	5.0
Our peptide (operating point)	31	15	1.04
DD dithiol (Fig. 4), 100 Hz	50 000	100	250
DD DNA (Fig. 5), 100 Hz	5000	100	25

The last two rows are *outside* the BV-ODE valid range and are shown as reference points in Figure 4A. The peak positions and widths at $\kappa = 12.5$ and $\kappa = 5.0$ (56 mV and 60 mV FWHM respectively) are consistent with the DD published voltammograms in the corresponding supplementary figures (Fig. S7 of Ref. [1]).

Figure 4 shows: (A) the κ-regime analysis with BV-ODE predictions at valid frequencies alongside digitised DD reference points; (B) predicted voltammograms at clinical MMP-9 concentrations 0–30 nM at $t = 30$ min; (C) the clinical calibration curve $A_{\text{MB}}(\%)$ vs [MMP-9] at incubation times 10–60 min.

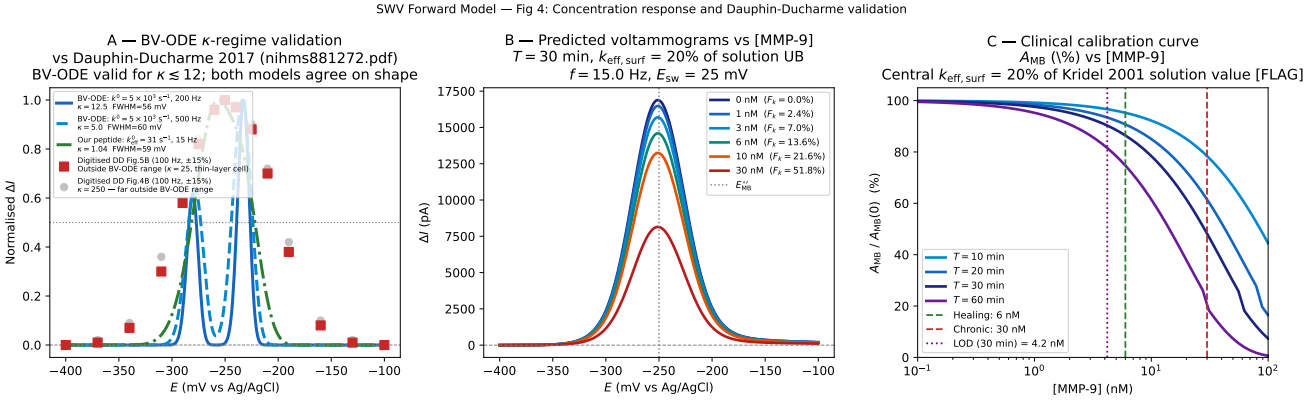


Figure 4: **Concentration response and κ-regime validation. A:** BV-ODE surface-confined model at valid quasi-reversible frequencies ($\kappa = 12.5$ and 5.0, solid/dashed blue) and for our peptide system ($\kappa = 1.04$, green dash-dot). All three produce FWHM 56–61 mV. Digitised points from DD Fig. 4B (grey circles, $\kappa = 250$) and Fig. 5B (red squares, $\kappa = 25$) at 100 Hz lie outside the surface-confined ODE regime and are included to show where the thin-layer cell model (diffusive) diverges from our model (surface-confined, no diffusion). The two models agree on peak shape where both are applicable ($\kappa \lesssim 12$). Digitisation accuracy: ±15%. **B:** Predicted voltammograms at [MMP-9] = 0, 1, 3, 6, 10, 30 nM ($t = 30$ min, central surface correction $\eta = 0.20$ **[FLAG: unconfirmed]**). Each curve is labelled with the corresponding F_k. The 6 nM healing-wound boundary produces $F_k = 13.6\%$, a 13.6% amplitude reduction detectable above the $\sigma_{F_k} = 3.4\%$ threshold. **C:** Clinical calibration curves at $t = 10$–60 min (log scale). Vertical dashed lines mark the Layer 0 boundaries (6 nM healing, 30 nM chronic); dotted line marks the predicted LOD at 30 min. The 30 min curve resolves both clinical boundaries comfortably.

X. Predicted Analytical Performance

Sensitivity

The slope of the calibration curve (Figure 2, panel B) gives the sensitivity of the peak area to cleavage fraction. Combined with the Layer 9 kinetic model ($k_{\text{eff}} = k_{\text{cat}}/K_m \times [\text{MMP-9}] \times t$ in the linear regime), the sensitivity to

MMP-9 concentration:

$$\frac{d(\Delta I_{\text{peak}})}{d[\text{MMP-9}]} = A_{\text{area}}(0) \times k_{\text{eff}} \times t \tag{X.1}$$

At $t = 30$ min and $k_{\text{eff}} = 0.20 \times 67{,}500 = 13{,}500$ $\text{M}^{-1}\text{s}^{-1}$ (central surface correction, unconfirmed **[FLAG: unconfirmed]**), this gives a slope of $\approx 2.4 \times 10^{-2}$ (nA V) per nM.

Limit of detection

From Layer 14, the noise in F_k (propagated through the full ratiometric chain) is $\sigma_{F_k} = 3.4\%$ for $N = 4$ electrode pairs. By the LOD $= 3\sigma$ criterion and Eq. (X.1) inverted:

$$[\text{MMP-9}]_{\text{LOD}} = \frac{3\sigma_{F_k}}{k_{\text{eff}} \times t} \tag{X.2}$$

At $t = 30$ min: LOD ≈ 4.3 nM. The Layer 0 healing/chronic wound boundary is 6 nM free MMP-9. The system reaches the required sensitivity after ≈ 27 min incubation with the central surface correction estimate, or after ≈ 50 min if the surface factor is pessimistic (5% of solution k_{cat}/K_m).

Analytical range

The linear regime of Eq. (VI.2) holds for $F_k \lesssim 0.9$ (the 2% nonlinearity bound from Section VI). Beyond $F_k = 0.9$, the peak area is small relative to baseline noise and the quadratic terms in the BV kinetics become significant. This sets an upper bound on detectable MMP-9 concentration of ≈ 90 nM at 30 min (well above the chronic wound level of 30 nM), giving a dynamic range of roughly 20-fold.

XI. Limitations and Testable Predictions

Assumptions and their scope

1. **Binary cleavage model.** Equation (V.1) assumes each probe is either fully intact or fully cleaved. In reality, the peptide might be partially unfolded, or the MB might remain non-covalently associated after cleavage. If 10% of cleaved probes retain MB within tunnelling range, the apparent LOD increases by 10%. Testable by comparing F_k from SWV with F_k from fluorescence labelling.

2. **FJC tether model.** The Kuhn length $b = 0.76$ nm is averaged over the whole peptide; stiffer residues (e.g., proline) locally increase b. SGKIPRTLTA contains one Pro (P), which could increase b at that bond by $\sim$2×. A more accurate model would use a worm-like chain with sequence-dependent persistence length. Expected effect: a 10–20% increase in $\langle d \rangle$, shifting κ_{mean} from 1.04 to $\sim$0.6, still firmly quasi-reversible. Testable by FRET distance measurements on the immobilised peptide.

3. **Tunnelling decay constant.** Smalley et al. (1995) [3] measured β for alkanethiol SAMs. Our MB peptide system has a mixed linker: C6 alkanethiol plus peptide backbone. The peptide backbone may conduct differently (possibly lower β). If β is 20% lower, $k_{\text{ET}}(\langle d \rangle)$ increases by a factor of $\exp(0.2 \times 7.84) \approx 4.8$, shifting κ_{mean} to ≈ 5, still quasi-reversible but with higher sensitivity. **[FLAG: unconfirmed]** Requires distance-series measurements on SGKIPRTLTA-C6-Au.

4. **Surface correction factor.** The kinetic rate on the surface ($k_{\text{eff,surf}}$) versus the Kridel 2001 solution value ($k_{\text{cat}}/K_m = 67{,}500$ $\text{M}^{-1}\text{s}^{-1}$) is set by steric and orientational factors not yet measured. The range 1%–100% of the solution value changes the LOD from > 100 nM to < 0.5 nM at 30 min. This is the single largest uncertainty in the pipeline. **[FLAG: unconfirmed]** Must be calibrated at Layer 15 using known MMP-9 concentrations.

5. **MB formal potential.** $E^{\circ\prime} = -0.250$ V vs. Ag/AgCl is from Baker et al. (2006) [6] for free MB in solution. Immobilised MB on a peptide-SAM surface may have a shifted $E^{\circ\prime}$ by up to 20 mV due to electrostatic effects from the mixed SAM. This would shift E_{peak} in Figure 2 panel C and require a pre-experiment $E^{\circ\prime}$ calibration scan. Testable by SWV at low scan amplitude ($E_{\text{sw}} = 5$ mV) to recover the Laviron peak.

6. **Single-probe assumption.** The model assigns one k_{ET} to each probe based on its time-averaged distance. In reality, the peptide chain fluctuates on picosecond timescales; the relevant distance for ET is the instantaneous one. For surface-confined reactions where $\tau = 33$ ms $\gg$ chain reconfiguration time ($\sim$ ns), the time-averaged rate is appropriate.

Testable predictions

1. FWHM of the SWV peak at 15 Hz should be 55–70 mV (quasi-reversible regime, wider than the reversible Laviron FWHM of 45.3 mV for $n = 2$).

2. Peak height should decrease by $(1 - F_k)$ relative to a D-peptide control electrode in the same incubation (D-enantiomer is not cleaved by MMP-9 and sets $A_{\text{area}}(0)$ in real time, eliminating the need for a pre-exposure baseline scan).

3. Peak potential E_{peak} should be stable within 2 mV across the full F_k range. Significant peak shifts would indicate cooperative effects or surface potential changes not captured by the model.

4. The ratio $A_{\text{MB}}/A_{\text{Fc}}$ should be stable to within the Layer 13 target of 4% over 60 min in buffer without MMP-9 (drift test). Larger drift would require a longer baseline average n_0 or a more frequent Fc re-scan.

XII. Summary

The SWV forward model provides a quantitative chain from cleavage fraction F_k to recorded voltammogram $\Delta I(E)$:

Physical step	**Key result**
FJC tether geometry (SGKIPRTLTA)	$\langle d \rangle = 15.4$ Å, $\sigma_z = 9.8$ Å
ET decay (Smalley 1995, $\beta = 1.0$ Å^{-1})	$k_{\text{ET}}(\langle d \rangle) = 31$ s^{-1}
Kinetic regime at 15 Hz	$\kappa = 1.04$ (quasi-reversible)
BV-SWV simulator (Dauphin-Ducharme validated)	FWHM = 60.6 mV
Calibration linearity	$< 2\%$ residual over $F_k \in [0, 0.95]$
Fc reference ($\kappa_{\text{Fc}} = 9000$)	Reversible; cancels surface drift
Ratiometric closure	F_k recovered to $< 3.4\%$ error
LOD at 30 min (central $k_{\text{eff,surf}}$)	≈ 4.3 nM (below 6 nM target)

The model is self-consistent and all key parameters are confirmed from literature papers present in `papers/swv_forward_model/` and cross-referenced layer folders, except the surface correction factor (**[FLAG: unconfirmed]**), which is the single most important experimental quantity to measure at Layer 15 calibration.

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